Science

Class 5 (English Medium)

Solved Exercises — Sindh Textbook Board

10 lessons — 147 questions

Classification of Living Things

Review

Question: What is classification? Why do scientists classify living things?

Answer: Classification means dividing living things into groups on the basis of their similar characteristics. Scientists classify living things so that millions of kinds of living things can be identified and understood easily, because it is very difficult to remember every living thing separately. Classification makes the characteristics of each living thing and its relation with other living things clear.

Question: Differentiate between vertebrates and invertebrates with examples.

Answer: Animals which have a backbone in their body are called vertebrates, for example fish, frog, snake, pigeon and cat. Animals which do not have a backbone are called invertebrates, for example insects, prawn, butterfly and earthworm. The number of invertebrates in the world is much greater than that of vertebrates.

Question: A whale looks like a fish. Why is it grouped with bats in mammals?

Answer: A whale looks like a fish but it is not a fish. A whale comes to the surface of water to breathe air, feeds its babies with milk and has warm blood. A bat also has the same characteristics. So both the whale and the bat are placed in the group of mammals, because classification is based on real characteristics and not on outward appearance.

Question: Identify the vertebrates and invertebrates from the following: octopus, snake, sparrow, earthworm, gold fish, frog, lizard, butterfly.

Answer: Vertebrates: snake, sparrow, gold fish, frog and lizard — all of them have a backbone. Invertebrates: octopus, earthworm and butterfly — they do not have a backbone. The identification is based only on the presence or absence of a backbone.

- ✗ True or false? All invertebrates live in water.
- ✓ True or false? Insects are invertebrates but they have hard outer coverings.
- ✗ True or false? Vertebrates are more in number than invertebrates.
- ✗ True or false? Animals which cannot make their own food are called producers.
- ✗ True or false? Green plants are called producers because they produce seeds.

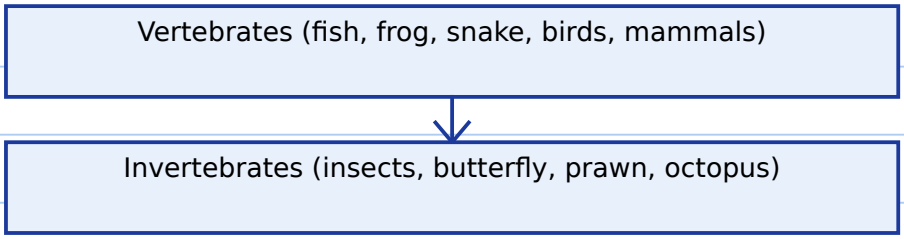
Question: Classify the following animals according to their eating habits: eagle, cow, crow, lion, goat, wall lizard, sparrow.

Answer: Carnivores (meat eaters): eagle, lion and wall lizard — they eat the flesh of other animals. Herbivores (plant eaters): cow and goat — they eat grass and plants. Omnivores (eat both): crow and sparrow — they eat seeds and plants as well as small insects.

Question: Activity: Make flash cards of twenty animals and divide them first into vertebrates and invertebrates, then into plant eaters, meat eaters or both eaters.

Answer: (Activity — model answer) Make cards with names and pictures of animals. Model division: Vertebrates — fish, frog, snake, pigeon, cat, cow, lion, eagle, sparrow, lizard; Invertebrates — butterfly, insect, prawn, earthworm, snail, octopus, housefly, crab, spider. Then by eating: plant eaters — cow, goat; meat eaters — lion, eagle; both eaters — crow, sparrow.

Classification of Animals



Microorganisms

Review

Question: What are microorganisms? Name the different types of microorganisms.

Answer: The very tiny living things which cannot be seen with the naked eye and can be seen only with a microscope are called microorganisms. Their main types are five: bacteria, virus, fungi, algae and protozoa. They can be found everywhere — in air, water, soil and inside and outside our body.

Question: Microorganisms constitute the largest number of living things on earth. Justify this statement.

Answer: Microorganisms are present everywhere — in air, in every drop of water, in every handful of soil and on the bodies of living things. They multiply very fast and one of them can become millions in a short time. Therefore their total number on the earth is much greater than all the other living things, although we cannot see them with our eyes.

Question: Suggest two diseases each which spread through air and water.

Answer: Diseases which spread through air: common cold and flu (influenza) — they spread in the air by the coughing and sneezing of a sick person. Diseases which spread through water: cholera and typhoid — they are caused by drinking polluted water. To stay safe from such diseases, drinking clean water and keeping cleanliness is necessary.

Question: How are microorganisms helpful to us in the environment?

Answer: Many microorganisms are friends of the environment. They decompose dead plants, animals and waste, which keeps the environment

clean and returns food (nutrients) to the soil. Some bacteria make the soil fertile and some are used in making our foods, such as yogurt and bread. So all microorganisms are not harmful.

Question: Sort out the following fungal diseases into diseases of plants and diseases of animals: 1. Athlete's foot 2. Wheat rust 3. Ringworm 4. Smut of maize.

Answer: Diseases of plants: wheat rust — it damages the wheat crop; smut of maize — it makes black patches on the maize plant. Diseases of human beings: athlete's foot — itching and cracking between the toes; ringworm — round red patches on the skin. All these diseases are caused by fungi.


True or false? All microorganisms are harmful.


True or false? Fungal diseases cannot be cured.


True or false? A mosquito is a microorganism.


True or false? Unhygienic conditions spread diseases.

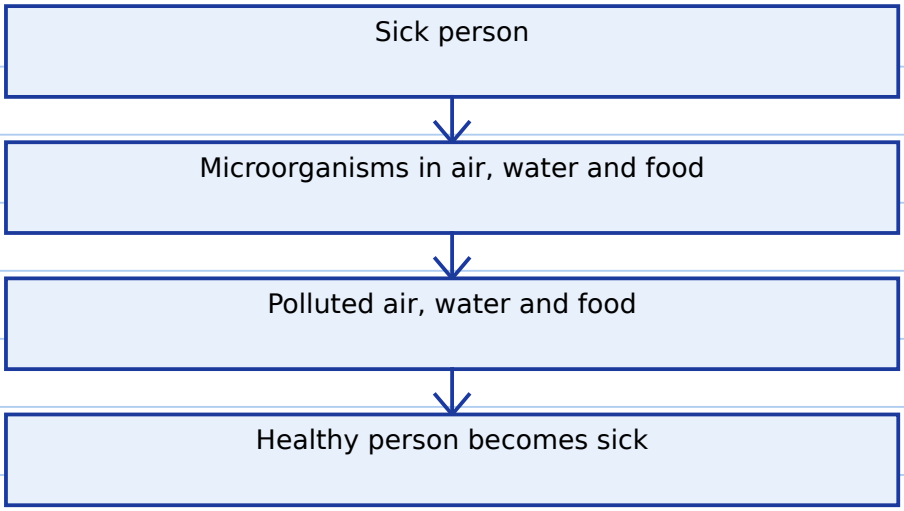
Question: Which microorganism is present in each of the following: (a) yogurt (b) common cold (c) malaria (d) athlete's foot?

Answer: (a) Yogurt contains bacteria, which change milk into yogurt and are useful. (b) Common cold is caused by a virus. (c) Malaria is caused by a protozoa (plasmodium), which spreads by mosquito bite. (d) Athlete's foot is caused by a fungus, especially in wet and unhygienic conditions.

Question: Activity: Find five diseases from the previous lessons, write their names and tell the cause of each disease.

Answer: (Activity — model answer) 1. Common cold — virus. 2. Flu (influenza) — virus. 3. Cholera — bacteria, spreads through polluted water. 4. Typhoid — bacteria, spreads through polluted water and food. 5. Malaria — protozoa, spreads through mosquito bite. Every disease is caused by some microorganism.

How Disease Spreads



Seeds: Their Structure and Germination

Review

Question: Name the parts of a seed and describe their functions.

Answer: A seed has three main parts. (1) Seed coat: it covers the seed from outside and protects it. (2) Cotyledons: food is stored inside them, which feeds the new plant at the time of germination; maize and wheat have one cotyledon and bean and gram have two cotyledons. (3) Embryo: it is the tiny young plant; its upper part (plumule) makes the shoot and leaves and its lower part (radicle) makes the root.

Question: What conditions are required for the germination of seeds?

Answer: Three things are necessary for the germination of seeds: water, air and suitable warmth. Water makes the seed soft and the young plant inside starts growing. Air (oxygen) is needed by the seed for breathing and getting energy. A seed germinates quickly in suitable warmth; in too much cold or too much heat it cannot germinate. Germination stops if any one of these conditions is missing.

Question: Name two plants that produce seeds without flowers.

Answer: Pine and fir are trees which do not make flowers. Their seeds are formed in cones, so they are called cone-bearing plants. Most other plants, such as mango and rose, make seeds from flowers, but these trees produce seeds without flowers.

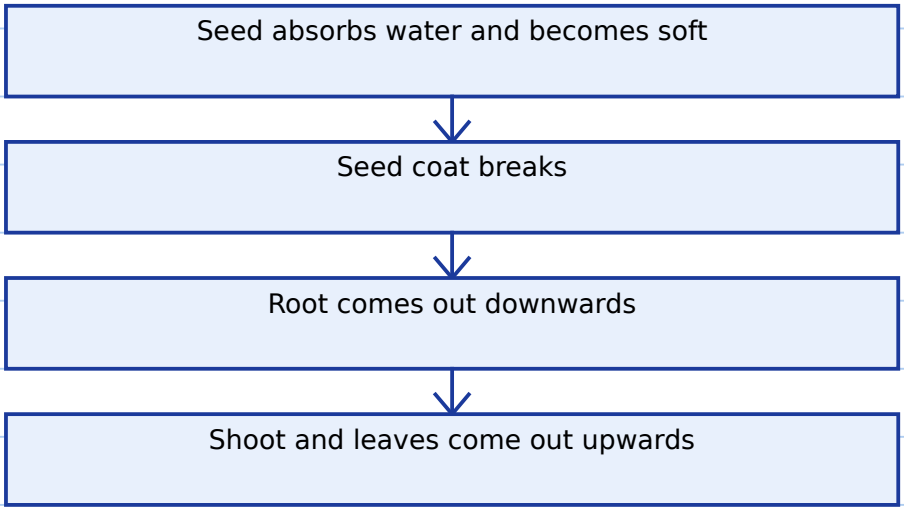
Match Column-1 with Column-2

Column-1	Column-2
1. Maize	Seeds in cones
2. Moss	Spores
3. Bean	Seeds with two cotyledons
4. Pine	Seed with one cotyledon

Question: Identify the seeds shown in the pictures (cotton seed, bean and wheat) and describe each of them.

Answer: (1) Cotton seed: it is small and has light hair (fibres) on it, which fly in air and spread the seed far and wide. (2) Bean seed: it is a big seed with two cotyledons (dicot) and food is stored inside it. (3) Wheat seed: it is a seed with one cotyledon (monocot) and is our important food. In this way seeds of different plants are different in shape and structure.

Germination of a Seed



Environmental Pollution

Review — Short Answers

Question: Define pollution and pollutant.

Answer: When the air, water or land around us becomes dirty and harmful, it is called pollution. The thing or substance which causes pollution is called a pollutant, for example smoke, sewage water, chemicals and plastic. Pollution is harmful for human beings, animals and plants alike.

Question: Why is smoke harmful to us?

Answer: Smoke contains harmful gases and tiny particles of dust. When we breathe air full of smoke, these enter our lungs and cause cough, breathing problems and diseases like asthma. Smoke makes the air dirty, makes it difficult to see, and forms a layer on the leaves of plants which stops their growth.

Question: Why should farmers not burn the remains of crops?

Answer: Burning the remains of crops produces a lot of smoke, which pollutes the air and makes people sick. The fire also kills the useful insects and microorganisms of the soil and reduces the fertility of the soil. Instead of burning, the remains should be mixed in the soil so that they rot and become natural manure which is useful for the crop.

Question: What are the beneficial effects of tree plantation on air?

Answer: Trees absorb carbon dioxide from the air and give out oxygen, which keeps the air clean and fresh. The leaves of trees stop the dust and smoke of the air and give shade and coolness in summer. When there are more trees, air pollution decreases and the environment stays pleasant. So we should plant more and more trees.

Question: How can we protect our water from pollution?

Answer: To protect water, dirt, garbage and sewage should not be thrown into rivers, canals and wells. The chemical water of factories should not be released into water without cleaning. Clothes and animals should not be washed near sources of drinking water, and cleanliness should be kept around hand pumps and wells. If everyone takes these precautions, our water will stay clean and free from diseases.

Long Answers

Question: Describe how germs reach drinking water.

Answer: Germs mix in water along with dirt. If open wells and hand pumps are near a dirty drain, dirty water seeps into the water. Rain water washes dirt into ponds and rivers. Germs also reach water through dirty hands and dirty pots. Drinking such polluted water causes diseases like cholera, typhoid and diarrhoea.

Question: Give the sources of noise pollution and its harmful effects.

Answer: The sources of noise pollution are: horns of vehicles, loud sound of loudspeakers, noise of factories and generators, sound of aeroplanes and construction work. Its harms are: too much noise reduces the hearing power, causes headache and lack of sleep. Students cannot concentrate on their studies and people remain irritable and sick. So unnecessary noise should be avoided.

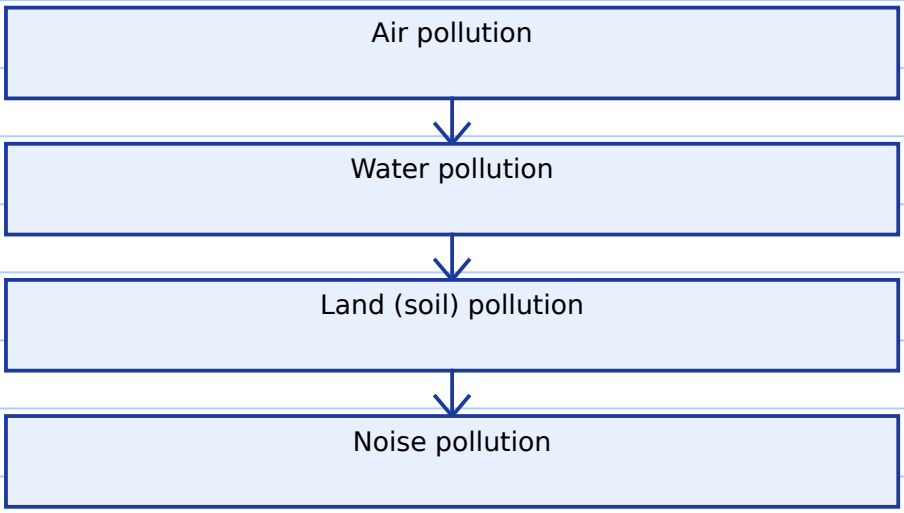
Question: How does cutting of trees affect the life of animals?

Answer: Trees are the homes of animals and birds and a source of their food. By cutting trees, birds lose their nests and animals lose their shelter. Many animals die due to lack of food or move to other places, and some kinds of animals are finished forever. Cutting trees also increases soil erosion and disturbs the whole food chain.

Question: Activity: Your neighbours release their waste water in the street instead of the drain. How will you convince them to stop doing this?

Answer: (Activity — model answer) I will politely and respectfully explain to the neighbours that waste water standing in the street produces mosquitoes and germs, spreads bad smell, and all the neighbours, especially children, can fall sick. I will request them to release the water in the drain or make a small soak pit near their house. If needed, I will sit together with the elders and people of the locality and get this solution adopted for everyone's health.

Types of Pollution



Matter and Changes in Its States

Review — MCQs

The melting point of ice is:

✓ (a) 0°C

(b) 20°C

(c) 32°C

(d) 100°C

At what temperature does water freeze to form ice?

(a) 100°C

✓ (b) 0°C

(c) 32°C

(d) 20°C

At what temperature does water boil?

(a) 0°C

(b) 37°C

✓ (c) 100°C

(d) 212°C

In gases, the particles of matter:

(a) do not move

(b) are close to each other

(c) move in two directions only

✓ (d) move freely in all directions

Which of the following is an example of sublimation?

(a) Water

(b) Milk

(c) Juice

✓ (d) Camphor

Solids change into liquid and then into gas on increasing temperature. What is the correct order?

(a) Melting, freezing

(b) Evaporation, condensation

✓ (c) Melting, evaporation

(d) Freezing, evaporation

Short Answers

Question: Define melting and freezing.

Answer: Melting is the process in which a solid changes into a liquid on heating, for example ice changing into water; it happens at a fixed temperature called the melting point. Freezing is the process in which a liquid changes into a solid on cooling, for example water changing into ice at 0°C. Both processes are opposite to each other.

Question: What happens if you keep a cube of ice for some time? Describe it.

Answer: The ice cube will absorb heat from the surrounding air and slowly melt into water. On getting heat, the particles of ice start moving more and leave their fixed positions and begin to flow. Until the ice melts completely, its temperature stays at 0°C, although its amount (mass) remains the same.

Question: It is the month of Ramzan. You buy ice from a shopkeeper but it melts before reaching home. How will you carry ice home without melting?

Answer: To save the ice from melting, I will wrap it in something which does not let heat pass in, such as a thick cloth, newspaper or sawdust. I will keep the ice in a thermos or a closed box, away from sunlight, and

take it home quickly. One big piece melts more slowly than small pieces, so I will buy the ice in one big piece.

Question: In the cooling curve of formation of ice, what does the part parallel to the time axis represent?

Answer: That straight (parallel) part represents the time when water is changing into ice, and during this time the temperature stays the same at 0°C. The liquid keeps losing heat, but its temperature does not fall, because that energy is used in changing the state. Only after the state has completely changed does the temperature start falling again.

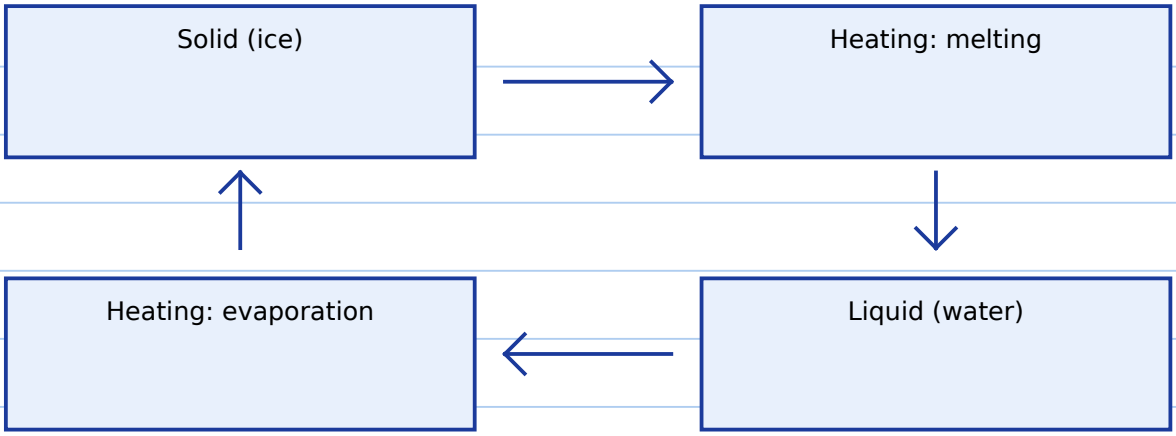
Question: Why do we feel cooling around an ice-cream cart?

Answer: Ice is kept in an ice-cream cart, and salt is mixed in it so that the ice cream stays frozen. While melting, the ice absorbs heat from its surroundings, which makes the air around the cart cool. That is why we feel coolness when we stand near the cart.

Question: Investigation: Make an ice lolly of your choice and write your observations on its melting.

Answer: (Investigation — model answer) Fill sweetened, coloured water in a mould and keep it in a freezer; in a few hours it will freeze into an ice lolly. When it is kept outside, it absorbs heat from the surroundings and starts melting, changing back into a sweet liquid. Observation: the lolly melts quickly in a warm place and slowly in a cool place; during melting, its temperature does not seem to rise at first — only its state changes.

Changes in the States of Matter



Force and Machines

Review — MCQs

Which of the following is NOT a simple machine?

(a) Lever

(b) Inclined plane

(c) Pulley

✓ (d) Computer

A wheel barrow is an example of which kind of lever?

✓ (a) Second kind

(b) First kind

(c) Third kind

(d) Fourth kind

Which machine has no moving parts?

(a) Screw

✓ (b) Inclined plane

(c) Wedge

(d) Lever

Which machine can we use for sliding a box up a slope?

(a) Pulley

✓ (b) Inclined plane

(c) Screw

(d) Lever

Question: How is an inclined plane helpful to us? (a) Pushing up a heavy load (b) Taking down a heavy load (c) Both (a) and (b) (d) None of these

Answer: (c) Both (a) and (b). On a sloping surface a heavy load can be

pushed up with less force and can also be brought down safely.

Which is an example of a second kind of lever?

(a) Nail cutter

(b) Fishing rod

(c) Pliers

(d) Claw hammer

Question: Define the following: lever, fulcrum, load, effort, pulley.

Answer: Lever: a rigid bar or plank which can turn around a fixed point and helps in lifting a load. Fulcrum: the fixed point on which the lever rests and turns. Load: the heavy object which is to be lifted or moved. Effort: the force applied by us to lift the load. Pulley: a small wheel with a groove around its rim; a rope passes over it, which makes lifting a load and changing the direction of force easier.

Question: Machines make our work easier but do not reduce the amount of work done. Prove this by an activity.

Answer: (Activity — model answer) First try to lift a heavy load straight up and note the force needed. Then slide the same load up an inclined plane or lift it with a lever — now much less force is needed, but the load has to travel a longer distance. Conclusion: a machine makes work easier by reducing the force, but the total work (force × distance) remains the same. So a machine does not reduce work; it only makes the way of doing it easier.

True or false? Load is the point where a lever is supported.

True or false? If a simple machine does not have a fulcrum, it is an inclined plane.

True or false? Winding roads in hilly areas are an example of an inclined plane.

True or false? A pulley is used to draw water from the well.

✗ True or false? Tongs is an example of a first kind of lever.

Fill in the blank: In _____ kind of lever, the effort is in the middle of the fulcrum and the load.

Answer: a third

Fill in the blank: In the first kind of lever, the fulcrum is in the middle.

Fill in the blank: A wheel barrow is a simple machine that is _____ kind of lever.

Answer: second

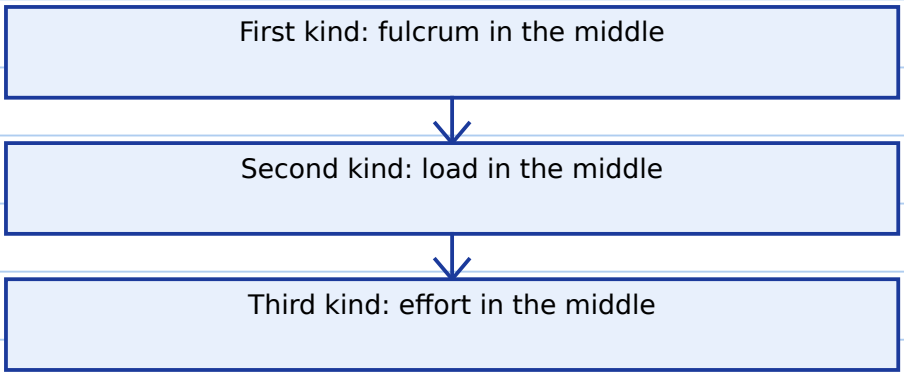
Fill in the blank: A nail cutter is an example of _____.

Answer: a lever (with a second kind lever the nail is cut with less effort).

Fill in the blank: Simple machines make our work _____.

Answer: easier by getting the work done with less force or in an easier way.

Three Kinds of Lever



Light and Its Properties

Review — Give Reasons

Question: Give a reason: A person can see the wall of his room but cannot see the particles of air.

Answer: We can see a thing when light reflects from it into our eyes. The wall is solid and a lot of light reflects from it to our eyes, so the wall is clearly visible. The particles of air are very small and transparent; light hardly reflects from them, so they cannot be seen.

Question: Give a reason: A shirt appears black.

Answer: A black surface absorbs all the colours of light and reflects no colour back. When no light reflects from the shirt into our eyes, it appears black to us. That is why a black shirt looks black.

Question: Give a reason: A rainbow always appears in the shape of an arc.

Answer: When sunlight passes through the tiny drops of rain, it bends (refracts) and splits into seven colours. Countless drops in the sky split the light in this way in the form of a big arc. That is why a rainbow always appears to us in the shape of a big coloured arc.

Question: Give a reason: When we go near a source of light, our shadow becomes bigger.

Answer: When an object is near a source of light, it blocks more rays of light, so its shadow is formed bigger. As we go away from the source, the object blocks fewer rays and the shadow becomes smaller. That is why the shadow looks bigger near the light.

Question: Give a reason: By looking through a prism we come to know that sunlight is made up of seven colours.

Answer: When white sunlight passes through a prism, it bends and splits into seven different colours, forming a band (spectrum): red, orange, yellow, green, sky blue, deep blue (indigo) and violet. This splitting proves that sunlight is actually a mixture of seven colours.

Question: Give a reason: We can see things around us.

Answer: The rays of a source of light, such as the Sun or a bulb, fall on objects and reflect from them into our eyes. When this reflected light reaches our eyes, we can see the objects. If there is no light, we cannot see anything.

Question: Give a reason: Nothing is visible in a dark room.

Answer: There is no light in a dark room, so there are no rays to reflect from the objects and no light reaches our eyes. Light is necessary for seeing, so we cannot see anything in the dark. As soon as a lamp is lit, the objects become visible.

✗ True or false? Only big objects make shadows.

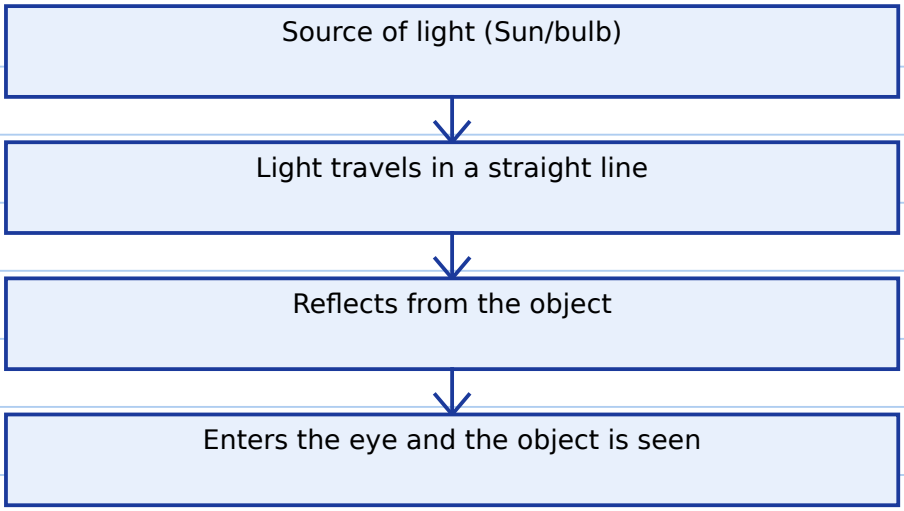
✗ True or false? Transparent materials allow all the light to pass through them.

✓ True or false? By noting the length of the shadow we can tell the time.

✓ True or false? Clear glass is a transparent material.

✗ True or false? There are ten colours in a rainbow.

Travel of Light and Seeing



Seven Colours of the Rainbow



Electricity and Magnetism

Review — True or False

✗ True or false? A static current flows in the wires of our homes.

✓ True or false? The flow of current in an electromagnet can be increased by increasing the number of turns of wire around an iron nail.

✓ True or false? Like charges repel each other.

✓ True or false? A compass always points towards the North.

✓ True or false? Hair can be electrified by friction.

Short Answers

Question: Give two examples of generating static electricity.

Answer: Example 1: Rubbing a plastic comb in dry hair produces static electricity, and the comb attracts small pieces of paper. Example 2: Rubbing a balloon on hair or a woollen cloth produces electricity in it and it sticks to the wall. In both examples the electricity is produced by friction and stays at one place.

Question: In the given pictures, the bulb glows in one circuit but not in the other. Why?

Answer: (Model answer) In the circuit in which the bulb glows, the wires are properly connected to both terminals of the cell and to the bulb, and the circuit is complete (closed), so current flows and lights the bulb. In the other circuit the circuit is open — either the switch is open, a wire is disconnected, or both wires are connected to the same terminal of the cell — so current cannot flow and the bulb does not glow.

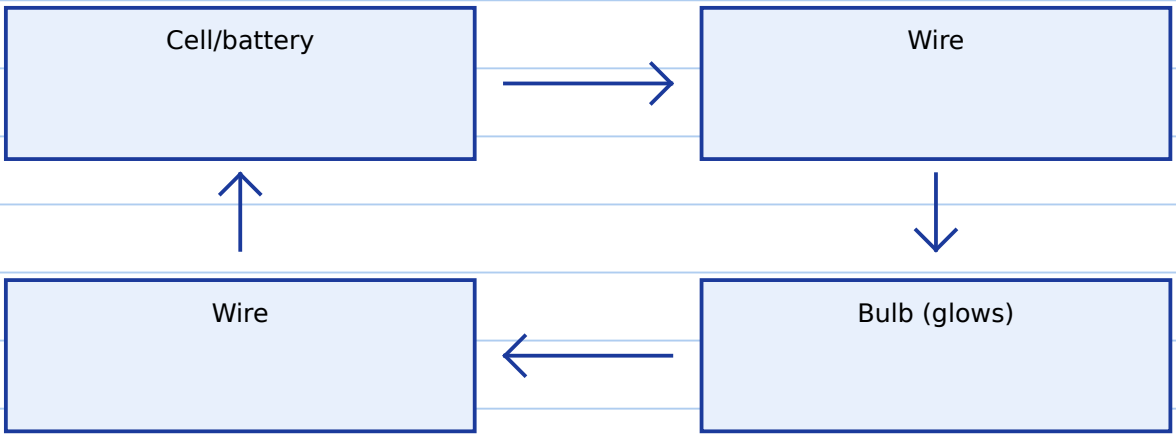
Question: Define an open circuit and a closed circuit.

Answer: Open circuit: a circuit in which the path of electricity is broken somewhere is called an open circuit; current does not flow in it and the bulb does not glow. Closed circuit: a circuit in which the path is complete and unbroken is called a closed circuit; current flows in it and the bulb glows. Opening and closing a switch makes the circuit open and closed.

Question: Draw and label a diagram of a simple circuit and a closed circuit.

Answer: A simple circuit has four parts: a cell or battery (source of electricity), wires (path of electricity), a bulb (which uses electricity) and a switch (to open and close the circuit). When the switch is closed, the circuit becomes complete; current comes out of the cell, reaches the bulb through the wires and returns to the cell, so the bulb glows. The parts of the circuit are shown in order in the diagram below.

A Simple Closed Electric Circuit



Soils

Review — Circle the Odd One Out

Question: Circle the odd one out and give the reason: (i) garden soil, sandy soil, clayey soil, clay pot (ii) soil, water, sunlight, earthworm (iii) humus, water, clay particles, sand particles (iv) stones, clay, sand, loam

Answer: (i) Clay pot — it is a thing made from soil, the other three are types of soil. (ii) Earthworm — it is a living thing, the other three are non-living. (iii) Water — water is not one of the layers of soil, the other three are parts or layers of soil. (iv) Loam — it is a complete type of soil made by mixing the others, the other three are parts found in soil.

✗ State true or false and correct the false statement: When you water the plants, all the water is absorbed.

✗ State true or false and correct the false statement: Plastic makes soil fertile.

✓ State true or false: Soil type 4 (a mixture of all the three soils) is the best for the growth of plants.

MCQs

Which type of soil is best for plant growth?

(a) Sandy soil

(b) Clayey soil

✓ (c) Loam — a mixture of sand, clay and humus

(d) Soil full of stones

Choose the dead organic matter from the following:

✓ (a) Dead ant, banana peel and fallen leaf

(b) Small stones and broken glass

(c) Plastic bag and sand

What is the order of the separation of soil into layers from top to bottom?

(a) Sand, silt, clay

✓ (b) Clay, silt, sand

(c) Clay, sand, silt

Short Answers

Question: What are the four components of soil?

Answer: The four main components of soil are: (1) mineral particles — sand, silt and clay, which are formed by the breaking of rocks; (2) humus — the matter of decayed plants and animals, which gives food to the soil; (3) water — which stays between the particles of soil and is available to plants; (4) air — which stays in the spaces between the particles and is necessary for roots and small organisms.

Question: Describe the activity of separation of soil into layers which shows that soil is a mixture of different sized particles.

Answer: Take soil up to half in a glass jar, fill it with water and shake it well. Then keep it without shaking for a few hours. You will see that the heavy and big particles (sand) settle at the very bottom, above it silt, above it fine clay, and decayed leaves and humus keep floating on the water. This proves that soil is really a mixture of particles of different sizes.

Question: If a soil sample is dry, how can it become wet quickly? What is this process called?

Answer: When water is poured on dry soil, the water goes down through the small spaces between the particles of the soil and the soil becomes wet. This process of water passing down through the soil is called percolation. Water percolates quickly through sandy soil, while this process is very slow in clayey soil.

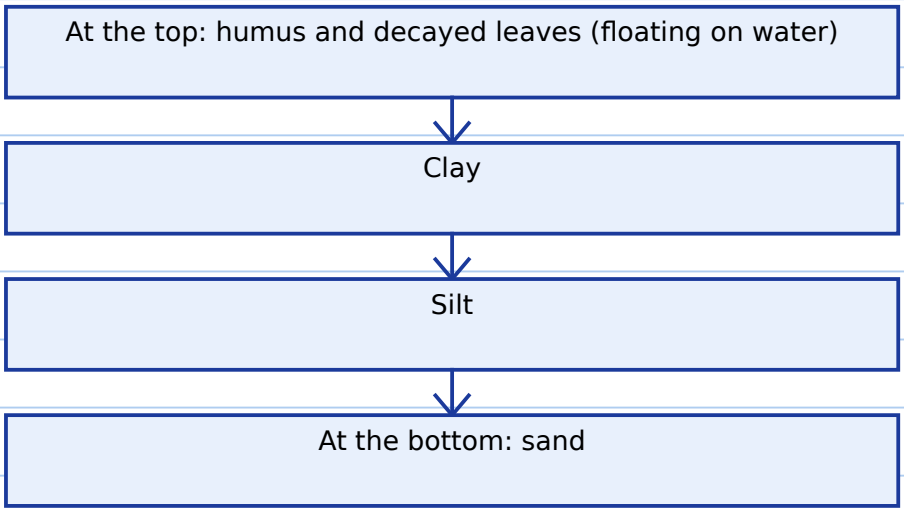
Question: Why is soil type 4 the best? Support your answer with an argument.

Answer: Soil type 4 (loam) is the best because it is a balanced mixture of sand, clay and humus. Sand manages air and drainage of water in it, clay holds water, and humus gives food to the plants. Therefore roots grow easily and plants grow healthy and fast.

Question: Clayey soil is sticky. Which soil is non-sticky?

Answer: Sandy soil is non-sticky. The particles of sand are big and there are large spaces between them, so water drains away quickly and it does not stick to the hands. On the contrary, the particles of clayey soil are very fine; it holds water and becomes soft and sticky.

Layers of Soil (in the Jar Experiment)



Our Solar System

Review — MCQs

The moon is a:

(a) star

(b) planet

✓ (c) satellite of the Earth

(d) comet

Planets move around the Sun in:

✓ (a) elliptical orbits

(b) circular orbits

(c) rectangular orbits

(d) triangular orbits

Which planet is known as the morning star?

(a) Mercury

✓ (b) Venus

(c) Earth

(d) Mars

Which is the smallest planet?

✓ (a) Mercury

(b) Mars

(c) Venus

(d) Earth

Which is the largest planet?

(a) Earth

(b) Saturn

(c) Jupiter

(d) Neptune

Which is the coldest planet?

(a) Mercury

(b) Earth

(c) Mars

✓ (d) Neptune

Which planet has the most number of moons?

(a) Jupiter

✓ (b) Saturn

(c) Uranus

(d) Earth

Neptune has _____ rings.

(a) Four

✓ (b) Six

(c) Five

(d) Ten

Which is the hottest planet?

(a) Mercury

(b) Mars

✓ (c) Venus

(d) Jupiter

Which of the following is a dwarf planet?

✓ (a) Pluto

(b) Neptune

(c) Mercury

(d) Earth

✓ True or false? The light of the Moon takes 2 seconds to reach the

Earth.

True or false? The light of the Sun takes 6 minutes to reach the Earth.

True or false? Venus is hot because of its thick atmosphere.

True or false? The Sun is denser than the Earth.

True or false? Mercury is the closest planet to the Sun, so it is the hottest planet.

Question: Which planet am I? (i) My clouds are dense, I am the hottest planet and I have a thick atmosphere. (ii) I am called red, I have a thin atmosphere and I have the largest volcano of the solar system. (iii) My atmosphere is mostly nitrogen and oxygen, I have water on my surface and life exists on me. (iv) I appear yellowish, I have the biggest family of 82 moons and my rings are made of ice and dust particles. (v) I am the biggest, my atmosphere is hydrogen and helium, I have a Great Red Spot and my moon Ganymede is the biggest moon.

Answer: (i) Venus — the hottest planet with dense clouds and a thick atmosphere. (ii) Mars — the red planet with a thin atmosphere and the largest volcano (Olympus Mons). (iii) Earth — the only planet with nitrogen and oxygen in its atmosphere, water and life. (iv) Saturn — the yellowish planet with 82 moons and rings of ice and dust particles. (v) Jupiter — the biggest planet, with the Great Red Spot and the biggest moon, Ganymede.

Question: Activity: Sort the following properties into those of Earth and those of Neptune: very cold planet; third planet from the Sun; six rings; takes about 165 years to revolve around the Sun; liquid water on the surface; deep blue in colour; atmosphere of nitrogen and oxygen; eighth planet from the Sun; only one moon.

Answer: (Activity — model answer) Earth: third planet from the Sun; liquid water on the surface; atmosphere of nitrogen and oxygen; only one moon. Neptune: very cold planet; six rings; takes about 165 years to revolve

around the Sun; deep blue in colour; eighth planet from the Sun. In this way every planet has its own separate properties.

Question: (i) Which planet has the most number of moons? (ii) Which planets have the least number of moons?

Answer: (i) Saturn has the most moons — 82 in total. (ii) Mercury has the least number of moons — it has no moon at all (zero). Venus also has no moon, while Earth has one and Mars has two moons.

Question: What is the function of a telescope?

Answer: A telescope is an instrument with which distant objects appear big and clear. Astronomers study the moon, planets, stars and other objects in space with a telescope. Through it we get a lot of information about the solar system and the universe, which cannot be obtained with the naked eye.

Question: What is the difference between asteroids and comets?

Answer: Asteroids are small rocky pieces which revolve around the Sun; most of them are found between Mars and Jupiter. A comet is a ball of ice, dust and rock; when it comes near the Sun, its ice melts and forms a long shining tail. Asteroids are made of rock and a comet is made of ice — this is the main difference.

Question: If you get a chance to visit a planet, which planet would you visit and why?

Answer: (Model answer) I would like to visit Mars, because it is the planet most similar to the Earth — the length of its day is like the Earth's and scientists are searching there for signs of water and life. I would go there and see Olympus Mons, the largest volcano of the solar system, and study its red soil. It would be an interesting and learning trip.

Question: Review: A thick blanket-like layer over Venus keeps it hot. Explain.

Answer: Venus has a thick atmosphere of carbon dioxide gas around it, which works like a blanket. This atmosphere lets the heat of the Sun come in, but does not let it go out — this is called the greenhouse effect. Therefore the heat stays inside and Venus becomes the hottest planet, although it is farther from the Sun than Mercury.

Order of Planets in the Solar System (from the Sun)

